

Competency Question Evaluation Matrix

Evaluation status definitions

Status	Meaning
Answerable	The required classes, properties, and restrictions are present at schema level.
Answerable with instance data	The schema supports the CQ, but concrete query results require populated individuals.
Partly answerable	The ontology contains part of the required structure, but the CQ needs a more precise interpretation, additional axioms, or instance data.
Answerable, but limited	The CQ is answerable, but the represented area has limited granularity.

Final CQ matrix

No.	Module	Competency Question	Required ontology elements	Evaluation result	Comment
CQ1	Stress Condition	Which stress-related conditions can be distinguished?	StressCondition, Stressor, NoStressCondition	Answerable	Covers the top-level distinction between stress-related and no-stress-related conditions.
CQ2	Stressor	What types of stressors can be described?	Stressor, AcuteStressor, ChronicStressor	Answerable	Covered by the stressor class hierarchy.
CQ3	Stressor	Which short-term stressors can occur in a stress-related setting?	AcuteStressor, PhysicalStressor, CognitiveStressor, SocioEvaluativeStressor	Answerable	Covers the acute stressors currently modelled.
CQ4	No-Stress Conditions	Which conditions describe baseline or recovery states?	NoStressCondition, BaselineCondition, RelaxationCondition	Answerable	Reflects the corrected model after the reasoner-based revision.
CQ5	Activity	Which activities or experimental phases can be described?	Activity, Baseline, PhysicalStressPhase, CognitiveStressPhase, SocioEvaluativePhase, RelaxationPhase	Answerable	Covered by the activity class hierarchy.
CQ6	Activity / Stress Condition	Which activity is linked to which stress condition?	Activity, hasStressCondition, StressCondition	Answerable	Central relation between activity phases and stress or no-stress conditions.
CQ7	Activity / Stressor	Which stressor is associated with a specific activity phase?	Stressor, hasAssociatedActivity, Activity	Answerable	Applies to stressors, not to baseline or relaxation conditions.
CQ8	Activity Metadata	What information can be used to describe an activity?	hasDurationSeconds, hasStartTimeSeconds, hasEndTimeSeconds, hasStartSample, hasEndSample, hasIntensity, hasDescription, hasReference	Answerable	Covers temporal and descriptive activity metadata.
CQ9	Person	Which roles can a person take in the model?	Person, Subject, Observer	Answerable	Distinguishes observed persons from observing persons.
CQ10	Subject	What information can be stored about a subject?	Subject, hasAge, hasGender, hasHeightCm, hasWeightKg	Answerable	Covers the current subject-related data properties.
CQ11	Sensor	Which sensor types can be used for stress-related measurements?	sosa:Sensor, PhysiologicalSensor, MotionSensor, AudioSensor, CameraSensor, HumanObserver	Answerable	Covers the main sensor categories.
CQ12	Physiological Sensor	Which physiological sensors are included?	PhysiologicalSensor, PPGSensor, HeartRateSensor, EDASensor, TemperatureSensor, SpO2Sensor	Answerable	Covered by the physiological sensor hierarchy.
CQ13	Motion Sensor	Which sensors can capture movement-related information?	MotionSensor, Accelerometer, GyroSensor, Magnetometer	Answerable	Covered by the motion sensor hierarchy.
CQ14	Platform	Which device or platform can be used as a measurement platform?	sosa:Platform, WearableDevice	Answerable	WearableDevice is represented as a SOSA platform.
CQ15	Sensor / Observable Property	Which properties can a heart rate sensor measure?	HeartRateSensor, sosa:observes, HeartRate, HeartRateVariability	Answerable	Covered by sensor-property restrictions.
CQ16	Sensor / Observable Property	What can be measured by an EDA sensor?	EDASensor, sosa:observes, ElectrodermalActivity	Answerable	Covered by sensor-property restriction.
CQ17	Sensor / Observable Property	What can be measured by a SpO2 sensor?	SpO2Sensor, sosa:observes, BloodOxygenSaturation	Answerable	Covered by sensor-property restriction.
CQ18	Sensor / Observable Property	What can be measured by a temperature sensor?	TemperatureSensor, sosa:observes, SkinTemperature	Answerable	Covered by sensor-property restriction.
CQ19	Sensor / Observable Property	Which acceleration-related properties can be captured by an accelerometer?	Accelerometer, sosa:observes, AccelerationX, AccelerationY, AccelerationZ	Answerable	Covered by accelerometer restrictions.
CQ20	Observable Property	Which categories of measurable properties are distinguished?	sosa:ObservableProperty, PhysiologicalProperty, BehavioralProperty, EnvironmentalProperty	Answerable	Covers the main observable property categories.
CQ21	Physiological Property	Which physiological properties can be measured?	PhysiologicalProperty, HeartRate, HeartRateVariability, ElectrodermalActivity, SkinTemperature, BloodOxygenSaturation, AccelerationX, AccelerationY, AccelerationZ	Answerable	Acceleration is currently modelled under physiological properties, although it can also be interpreted as movement-related context.
CQ22	Behavioral Property	Which behavioral indicators can be described?	BehavioralProperty, Gesture	Answerable, but limited	The structure exists, but behavioral indicators are only modelled at low granularity.

No.	Module	Competency Question	Required ontology elements	Evaluation result	Comment
CQ23	Environmental Property	Which environmental factors can be captured?	EnvironmentalProperty, Humidity, OutsideTemperature	Answerable	Covered by environmental property classes.
CQ24	Environment	What environmental information can be assigned to an environment?	Humidity, OutsideTemperature, hasHumidity, hasOutsideTemperature	Partly answerable	The values are attached to Humidity and OutsideTemperature, not directly to Environment.
CQ25	Record	What metadata can be stored for a physiological record?	PhysiologicalRecord, hasFileName, hasFileFormat, hasNumberOfChannels, hasNumberOfSamples, hasAnnotationRecord	Answerable	Covers record-level metadata and annotation linkage.
CQ26	Record / Channel	How can a physiological record be connected to its signal channels?	PhysiologicalRecord, hasSignalChannel, SignalChannel, isSignalChannelOf	Answerable	Covers both directions through inverse properties.
CQ27	Signal Channel	What information can be stored for a signal channel?	SignalChannel, hasChannelName, hasSamplingFrequencyHz, hasUnit	Answerable	Covers channel-level metadata.
CQ28	Signal Channel / Observable Property	How can a signal channel indicate what it measures?	SignalChannel, representsObservableProperty, sosa:ObservableProperty	Answerable	Links a data channel to its semantic observable property.
CQ29	Observation	Which types of observations can be described?	sosa:Observation, TimeSeriesObservation	Answerable	Time-series observations specialize SOSA observations.
CQ30	Time-Series Observation	What temporal information can be assigned to a time-series observation?	TimeSeriesObservation, hasTimestamp, hasWindowStartSeconds, hasWindowEndSeconds	Answerable	Timestamp is defined for TimeSeriesObservation; window start and end are defined for sosa:Observation.
CQ31	Observation / Source Data	How can an observation be traced back to its source data?	sosa:Observation, hasSourceRecord, PhysiologicalRecord, hasSourceChannel, SignalChannel	Answerable	Supports traceability from observations to records and channels.
CQ32	Observation / Activity	How can an observation be assigned to the activity during which it was made?	sosa:Observation, observedDuringActivity, Activity	Answerable with instance data	The property exists; concrete answers require observation and activity individuals.
CQ33	Activity / Observation	How can measurements be linked back to the activity in which they occurred?	Activity, hasObservation, sosa:Observation	Answerable with instance data	The inverse relation exists; concrete results require populated individuals.
CQ34	Annotation	What information can be stored in an annotation?	AnnotationRecord, hasNote, hasAnnotationTimeStamp	Answerable	Covers note text and annotation timestamp.
CQ35	Annotation / Activity	How can an annotation be assigned to an activity or phase?	AnnotationRecord, hasPhase, Activity	Answerable with instance data	The property exists; concrete answers require annotation and activity individuals.
CQ36	Observer / Annotation	How can an observer be connected to an annotation?	Observer, makesAnAnnotation, AnnotationRecord	Answerable with instance data	The property exists; concrete answers require observer and annotation individuals.
CQ37	Observation Source	How can a human observer be identified as the source of an observation?	sosa:Observation, madeByObserver, Observer, HumanObserver	Answerable with instance data	Supports non-sensor-based observation sources.
CQ38	Biomarker	How can biomarker assays be described?	BiomarkerAssay, hasAssayTimestamp	Answerable	Covers the biomarker assay class and timestamp metadata.
CQ39	Biomarker / Observation	How can an observation be linked to a biomarker assay?	sosa:Observation, madeByBiomarkerAssay, BiomarkerAssay	Answerable with instance data	The schema supports linking observations to biomarker assays.
CQ40	Biomarker Timestamp	When can the time of a biomarker assay be documented?	BiomarkerAssay, hasAssayTimestamp, xsd:dateTime	Answerable	Covered by the biomarker timestamp data property.